EAP: Architecture Specification and Prototype

> Designed to redefine the reading experience, Liberato brings together technology, accessibility, and community to create a modern space for book lovers. The platform allnnovations readers to discover, purchase, and review books with ease. By giving costumers a seamless shopping experience, Liberato empowers readers, authors, and publishers to connect through a shared passion for literature.

A7: Web Resources Specification

> This artifact describes the web API to be developed for Literato, detailing the required resources, their properties, and JSON responses. This API includes create, read, update, and delete operations (where applicable) for each resource.

1. Overview

> Identification and overview of the modules that will be part of the application.

Identifier	Module	Description
M01	Authentication	Web resources for user authentication and registration, including login, logout and user sign-up.
M02	Profile	Web resources for accessing user information: profile page, edit profile, password recovery and account settings.
M03	Catalog	Web resources for browsing and viewing books: listings, detail pages, categories, genre and pagination.
M04	Shopping Cart	Web resources to add/update/remove items; session cart for guests and DB-backed cart for authenticated users.
M05	Checkout & Orders	Web resources for checkout, order creation, order confirmation and order history and stock checks.
M06	Payments	Web resources for payment handling, supports Card, MBWay and PayPal.
M07	Search & Discovery	Web resources for Exact Match and Full-Text Search, filters, ordering and faceted/multi-attribute queries.
M08	Reviews & Ratings	Web resources to view, write, edit and remove product reviews, including moderation/reporting capabilities.
M09	Notifications	Web resources for user notifications and alerts (order status, wishlist availability, price changes); notification list and mark-as-read actions.
M10	Administration	Admin web resources to manage users, products, orders, stock, categories and platform moderation.
M11	Seller Tools	Web resources for sellers to manage products, stock and sales statistics.
M12	Home & Platform Info	Web resources for the landing page, About, Contact and platform metadata used across the site.

2. Permissions

> The permissions used by each module, necessary to access its data and features.

Identifier	Name	Description
PUB	Public	Unauthenticated user (guest) - can view public pages and access login/register.
USR	User	Authenticated user - can access profile, cart and purchase fInnovations (prototype), and other user-only features.
BLK	Blocked	Blocked user - restricted from normal user actions.

Identifier	Name	Description
OWN	Owner	Owner of a resource (e.g., a user owns their profile or a user owns their content) - used for owner-only actions.
ADM	Administrator	Admin of the system - full access to `/admin` and management operations.

3. OpenAPI Specification

> OpenAPI specification in YAML format to describe the vertical prototype's web resources.

> Full file available at `a7_openapi.yaml` in the associated repository.

```yaml
openapi: 3.0.0
info:
  title: Liberato API
  description: |
    OpenAPI specification for the Liberato online bookstore.
    This documents the public catalog, shopping cart, user profile, and admin endpoints.
  version: 1.0.0
servers:
  - url: http://127.0.0.1:8001/
    description: Local Development Server

paths:
...
```

A8: Vertical prototype

> The Vertical Prototype implements the project's highest-priority user stories to validate the chosen architecture and to provide hands-on familiarity with the technologies used in the project. This artifact lists the implemented user stories, maps each implemented story to the exact web resources (routes, controller, views) and the steps necessary to test the prototype.

1. Implemented Features

1.1 Implemented User Stories

> User stories that were implemented in the prototype.

1.1.1 User (unauthenticated)

Identifier	Name	Priority	Responsible	Description
US01	Registration	High	João Santos	As a visitor, I want to register a new account, so that I can start using the system.
US02	Login	High	João Santos	As a user, I want to log in to my account, so that I can securely access and end my sessions.
US03	View & Search Products	High	João Santos	As a user, I want to view and search products, so that I can browse the store.
US04	Exact Match Search	High	João Santos	As a user, I want to search with exact matches, so that I can quickly find specific results.
US05	Full-text Search	High	Dinis Pinto	As a user, I want to use full-text search, so that I can find results by keywords.

Identifier	Name	Priority	Responsible	Description
US06	Add item to Cart	High	Iara Brás	As a user, I want to add items to my cart, so that I can buy my purchases.
US07	Manage Cart	High	Iara Brás	As a user, I want to update or remove items in my cart, so that I can manage my purchases.
US08	View Product Details	High	João Santos	As a user, I want to view detailed information about a product, so that I can make informed purchase decisions.
US09	View Product Reviews	Medium	João Santos	As a user, I want to view product reviews, so that I can evaluate others' experiences before buying.
US11	Search Filters (category)	Medium	João Santos	As a user, I want to apply filters to my search, so that I can narrow down results easily.
US13	Browse Product Categories	Medium	João Santos	As a user, I want to browse product categories, so that I can easily find types of products I'm interested in.
US15	Placeholders in Form Inputs	Innovation	Ricardo Fernandes	As a user, I want to see placeholders in form inputs, so that I understand what information to enter.

1.1.2 User

Identifier	Name	Priority	Responsible	Description
US10	Search over Multiple Attributes	Medium	Dinis Pinto	As a user, I want to search by multiple attributes, so that I can refine results.

1.1.3 Authenticated user

Identifier	Name	Priority	Responsible	Description
US19	View Profile	High	João Santos	As an authenticated user, I want to view my profile information, so that I can check my personal data.
US20	Edit Profile	High	João Santos	As an authenticated user, I want to edit my profile, so that I can update my information when needed.
US21	Checkout	High	Iara Brás	As an authenticated user, I want to proceed to checkout, so that I can complete my purchase securely and efficiently.
US26	Upload/Update Profile Picture	Medium	João Santos	As an authenticated user, I want to upload or update my profile picture, so that my account is personalised.
US32	Log Out	Medium	João Santos	As an authenticated user, I want to log out of my account, so that I can ensure my personal data and session remain secure when I leave the system.
US34	Manage Purchases	Medium	Ricardo Fernandes	As an authenticated user, I want to view my purchase history, so that I can track past orders.

1.2. Implemented Web Resources

> Identify the web resources that were implemented in the prototype.

> Module M01: Authentication & Profile

Web Resource Reference	URL
R01: Login Form	`/login` (GET)
R02: Login Action	`POST /login`
R03: Logout Action	`GET /logout`
R04: Register Form	`/register` (GET)
R05: Register Action	`POST /register`
R06: Profile (show/update)	`/profile` (GET / POST, auth required)

> Module M02: Catalog

Web Resource Reference	URL
R11: Catalog listing (paginated, search, category)	`/catalog` (GET, supports `?q=` and `?category=`)
R12: Book detail	`/catalog/{book}` (GET)

> Module M03: Shopping Cart

Web Resource Reference	URL
R21: Cart page (view)	`/cart` (GET)
R22: Add item to cart	`POST /cart/add`
R23: Buy action	`POST /cart/buy`
R24: Checkout page	`/cart/checkout` (GET)
R25: Complete purchase	`POST /cart/complete`
R26: Update cart quantities	`PUT/PATCH /cart/update`
R27: Remove item from cart	`POST /cart/remove`

Note: The prototype implements a hybrid cart strategy — guests use a session-based cart while authenticated users have a DB-backed cart stored in the `shopping_cart` table. On user login, the session cart is merged into the DB cart (`app/Providers/AppServiceProvider.php`), and controller actions keep the session in sync for the UI.

> Module M04: Home & Platform Information

Web Resource Reference	URL
R31: Home / landing page	`/` (GET)

2. Prototype

> Command to start the Docker image from the group's Container Registry. User credentials necessary to test all features. Link to the source code in the group's Git repository.

Start the docker image:

```
docker-compose build
```

```
docker-compose up
```

Outros:

```
php artisan storage:link  # abre a ligação para utilização de imagens
```

User Credentials:

Admin

- Email: admin1@liberato.com
- Password: password

Regular User

- Email: john@example.com
- Password: password

The code is available at: https://gitlab.up.pt/lbaw/lbaw2526/lbaw25134

Revision history

Changes made to the first submission:

1. ...
2. ...

GROUP25134, 26/11/2025

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Comments